Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

Farshid Mossaiby[a,*]

[a] *Department of Civil Engineering, University of Isfahan, 81744-73441, Isfahan, Iran*

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Research Highlights

- Weighted neural reads are expressed over valued measurable atom spaces.
- Exact top-k truncation has two bounds and an error-budget rule.
- CounterFact-500 exposes strong retrieval but weak paraphrase recall.
- Signed-read perturbation bounds quantify approximate cancellation failure.
- Five-seed matched routing exposes corpus-to-parameter confusion.

*Corresponding author.
Email address: mossaiby@eng.ui.ac.ir (Farshid Mossaiby)